Genomewide aggregated *trans*- effects analysis supports causal role in hypertension for immune system and lipid handling in endothelial cells

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Abstract

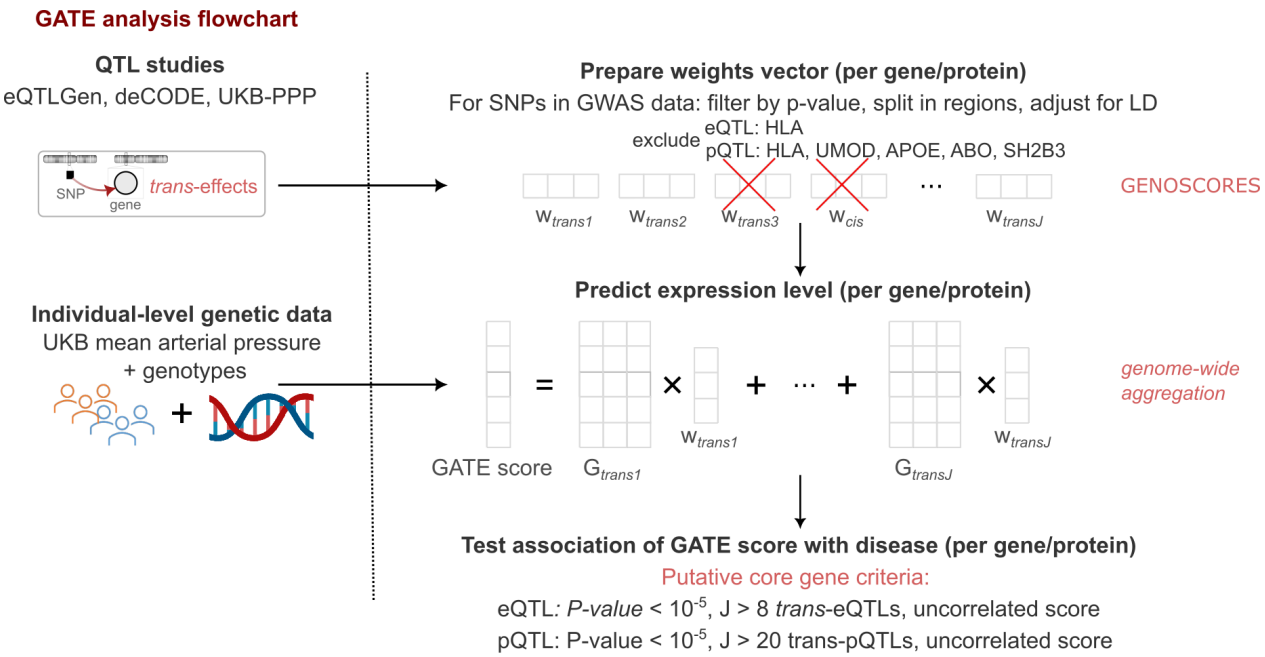
Background: Genome-wide association studies have identified thousands of genomic regions in which common variants affect blood pressure, but have yielded limited mechanistic insight. The "omnigenic" sparse effector hypothesis postulates that for a typical complex trait the weak polygenic *cis*-effects of common variants coalesce via *trans*-effects on expression of a set of core genes that directly influence the trait. The objective of this study was to identify core genes influencing blood pressure.

Methods: Using summary statistics for quantitative trait loci (QTLs) from eQTLGen Phase 1, deCODE proteomics and UK Biobank proteomics studies, we computed genomewide aggregated *trans*-effects (GATE) scores for XX genes in 215,881 UK Biobank participants of European ancestry aged less than 60 years and tested these scores for association with mean arterial pressure (MAP).

Results: MAP was associated with *trans*-effects on expression of several genes that are involved in handling of lipids by endothelial cells: *CD36*, *CD300LG*, *ADIPOQ* and *DSG2*. Raised blood pressure was inversely assocated with *trans*-effects on expression of a cluster of co-regulated genes that are expressed on B cells, including *BANK1*, *COBLL1*, *IL2RB* and *MEF2C*, and positively associated *trans*-effects on circulating butyrophilin 2A1, which mediates the response of $\gamma\delta$ T cells to intracellular phosphoantigens. MAP was inversely associated with *trans*-effects on circulating levels of interleukin-17 receptor B and positively associated with _trans–effects on circulating levels of interleukin-2 receptor B.

Conclusions: These results provide genetic support for causality of pathways that may be amenable to the development of new pharmacological interventions for hypertension: specifically lipid handling in endothelial cells, immune response to endogenous phosphoantigens and interleukin-mediated inflammatory signaling.

Graphical Abstract



Non-standard abbreviations and acronyms

CHARGE, Cohorts for Heart and Aging Research in Genomic Epidemiology; eGATE, eQTL-based Genome-wide Aggregated *Trans*-Effects score; eQTL, expression quantitative trait locus (loci); GATE, Genome-wide Aggregated *Trans* Effects; GWAS, genome-wide association study (studies); HDL, high-density lipoprotein; HLA, human leukocyte antigen; IgG, immunoglobulin G; LD, linkage disequilibrium; MAP, mean arterial pressure; MR, Mendelian randomization; NK, natural killer (cells); pGATE, pQTL-based Genome-wide Aggregated *Trans*-Effects score; pQTL, protein quantitative trait locus (loci); RAAS, renin-angiotensin-aldosterone system; SKAT-O, Sequence Kernel Association Test–Optimised; SNP, single nucleotide polymorphism; UKB, UK Biobank; UKB-PPP, UK Biobank Pharma Proteomics Project; VLDL, very low-density lipoprotein

Introduction

For blood pressure traits, large-scale genome-wide association studies (GWAS) have now identified on the order of two thousand common-variant loci[1]. However, these loci collectively explain only a small fraction of the heritability of blood pressure (around 1–3% of the phenotypic variance in individual-level studies), underscoring the highly polygenic architecture of hypertension and providing only limited direct insight into its underlying aetiology[2]. Most associated variants have small effect sizes, are non-coding, and map to broad genomic regions containing multiple candidate genes, making it challenging to pinpoint the specific genes and pathways that are truly causal and therapeutically tractable.

Recent advances in statistical genetics and the availability of large-scale genomic and molecular datasets have enabled the development of Genome-wide Aggregated *Trans* Effects (GATE) analysis, which aggregates genome-wide *trans*-acting genetic effects to prioritise "core genes" with pivotal effects in disease[3]. The approach has been successfully applied to oligogenic autoimmune disorders, where it has highlighted genes with clear pathogenic relevance and therapeutic potential[3–7]. Extending GATE to a highly polygenic trait such as hypertension offers an opportunity to translate diffuse GWAS signals into a more interpretable set of biologically central genes and pathways. In this study, we applied GATE analysis to UK Biobank (UKB) data to identify core genes and proteins causally implicated in hypertension. One specific objective was to investigate the role of inflammation driven by the innate and adaptive immune systems[8].

Methods

Data and resource availability

UK Biobank data are available on application via the UK Biobank access portal. Analysis code that generates this manuscript from summary-level data is provided in our GitHub repository. Summary-level,data is available on Zenodo. and summary statistics for associations between mean arterial pressure and all GATE scores are archived on Zenodo.

Study populations and phenotype definition

The UK Biobank cohort includes approximately 500,000 UK volunteers aged 40–69 years recruited between 2006 and 2010[9]. We used individual-level genotype and phenotype data from UKB as a target cohort, restricted to participants of European ancestry with genome-wide SNP data and valid blood pressure measurements, averaged over two assessments. Ethical approval for UKB was granted by the North West Centre for Research Ethics Committee (11/NW/0382). Written informed consent was obtained from all participants. The present study was conducted under UKB approval (application 23652).

Mean arterial pressure (MAP) was defined as two-thirds diastolic plus one-third systolic blood pressure. To reduce confounding and reverse causation, we restricted the main analysis to participants aged <60 years, in whom genetic effects on blood pressure tend to be stronger and age-related comorbidities and treatment patterns are less likely to confound associations[10]. We also excluded individuals with common risk factors for hypertension, such as chronic kidney disease (estimated glomerular filtration rate <60 mL/min/1.73 m^2) or type 1 diabetes[11–13]. When testing GATE scores derived from UKB-PPP pQTL summary statistics, individuals included in UKB-PPP were excluded from all association analyses to avoid overlap between the QTL and target datasets. We removed related individuals until no pairs had kinship coefficients > 0.05, then calculated genotypic principal components from the unrelated set. After all exclusions, 215,881 individuals remained in the dataset with typed or imputed genotypes for 9869162 SNPs. . For participants receiving antihypertensive therapy, untreated blood pressure was imputed by adding 15 mmHg to systolic and 10 mmHg to diastolic blood pressure, in accordance with CHARGE consortium recommendations[14].

Computation of GATE scores

GATE analysis is a two-stage procedure, as illustrated in the Graphical Abstract. In the first stage, we use publicly available regression coefficients (weights) for the association of genome-wide SNP data with gene expression or protein levels (quantitative trait loci or eQTL/pQTL GWAS summary statistics). In the second stage, we apply these weights to individual-level genotypes in the target dataset (UKB) to compute a locus-specific *trans* score for each region and a genome-wide aggregated *trans* (GATE) score for each transcript or protein. These scores then are tested for association with the mean arterial pressure (MAP) in the target dataset. Genes whose GATE scores are significantly associated with the phenotype ($P < 10^{-5}$), consist of effects of multiple unlinked *trans*-QTLs are prioritized as putative core genes (hereafter referred to as core genes). GATE analysis methods have been reported previously[3–7].

We used publicly available eQTL and pQTL GWAS summary statistics to derive SNP weight vectors for whole-blood transcripts and plasma proteins from three studies: eQTLGen Consortium Phase I (10,317 *trans*-eQTLs for 6,298 genes in 31,684 participants[15]), deCODE (4,719 proteins on the SomaLogic v4 panel in 35,559 participants[16]), and the Olink Explore panel in 54,219 UK Biobank participants (UKB-PPP, 2,923 proteins[17]). Quality control procedures and SNP–protein coefficient estimation have been reported previously[17–19]. 1975 SomaLogic aptamers with evidence of cross-reactivity with CFH were excluded from the deCODE data. The eQTL and pQTL datasets were used solely to estimate SNP weights and did not contribute phenotype information.

For each genomic region with at least one SNP associated with a transcript or protein at $P < 10^{-6}$, we defined a quantitative trait locus by grouping all SNPs associated with that transcript or protein at $P < 10{-}5$. A locus was classified as *cis* if its distance from the transcription start site of the encoding gene was < 50 kb, *cis*-x if 50 kb to 5 Mb, and *trans* if > 5 Mb. As in previous studies, to avoid confounding by large direct effects on disease, we excluded *trans*-eQTLs in the HLA region (25 to 34 Mb on chromosome 6) from GATE scores. As trans-effects on hypertension are weak and there are effect To account for the polygenicity of hypertension, for we additionally excluded QTLs in the *UMOD*, *APOE*, *ABO*, and *SH2B3* regions, which have pleiotropic effects on gene expression and circulating protein levels and also affect blood pressure. **include SH2B3 and UMOD SNPs as covariates, no need to exclude regions where we can control for the protein alleles**

To obtain linkage disequilibrium (LD)-adjusted SNP weights, we premultiplied the univariate regression coefficients by the inverse genotype covariance matrix in a reference panel of European ancestry. For each individual, the GATE score for a transcript (eQTL GATE score) or protein (pQTL GATE score) was then calculated as the dot product of the genotypes in the target cohort and these LD-adjusted weights at *trans*-QTLs, representing the genetically determined expression or protein level from all *trans*-QTLs. The locus-specific contributions are provided in Supplementary Data files. SNPs corresponding to *cis*-QTLs for the gene were aggregated separately into a *cis* score, independent from the GATE score.

To quantify the effective number of unlinked *trans*-QTLs contributing to each genome-wide *trans*-score, we calculated the Hill diversity index[20]. For each gene, the diversity index was computed from the variances of the K locus-specific *trans* scores as $2^{-\sum_{i=1}^{K} p_i \log_2 p_i}$. Here, $p_i = \sigma_i^2 / \sum_{j=1}^{K} \sigma_j^2$ and σ_i denotes the variance of the i-th locus-specific *trans* score. This index ranges from 1 (when one QTL substantially dominates variance) to K (when locus-specific

variances are approximately equal).

As the objective of this study was to identify core genes on which the *trans*-effects of many common variants coalesce to influence blood pressure, we focused the results shown in this study on genes for which the effective number of unlinked *trans*-QTLs was large. GATE score associations with such genes are more likely to be causal because as the effective number of trans-QTLs increases, the signal of a causal effect (QTL effects consistent in direction) becomes stronger in relation to the "noise" of pleiotropic QTL effects.

Testing for association with GATE scores

We tested associations between GATE scores and MAP in unrelated UKB participants of European ancestry, consistent with the European-ancestry restriction of the eQTL and pQTL studies used in the first step. For each score, we fitted a linear regression model with MAP as the continuous outcome and the GATE score as the predictor, adjusted for age, sex, body mass index, body fat percentage, self-reported ethnic background, alcohol consumption, autoimmune thyroid deficiency and 40 genotype principal components. The covariates were chosen based on a regression model seeking association to determine what were the main covariates in the UKB dataset that are associated with blood pressure variation. We adjusted for measures of adiposity because we were explicitly interested in genetic determinants of hypertension beyond those mediated by obesity. We also excluded individuals with autoimmune thyroid disease to reduce potential confounding by autoimmunity[21]. *Cis* score associations with MAP were tested separately.

Fitted values from this base model were then used to compute score tests for association with the *cis* score and the aggregated *trans* score for each transcript and each circulating protein. Score tests, based on the gradient and negative second derivative of the log-likelihood at the null, are a computationally efficient way to compute tests of association for many variables one at a time against the same null model. Log odds ratios reported in the tables were standardized by multiplying the corresponding regression coefficients by the standard deviation of the score, so that each coefficient can be interpreted as the log odds ratio associated with a one-standard deviation increase in the score.

For each putative core gene with at least 10 *trans*-QTLs, we tested for a dose-response relationship between the effect of the *trans*-QTLs on the expression of the gene or protein and the effect of those same *trans*-QTLs on blood pressure. This is equivalent to a 2-step Mendelian randomization analysis as described previously[5–7]. The method is similar to a recently described Bayesian method[22], but instead of reporting a posterior distribution we obtain the marginal likelihood of the causal effect parameter by dividing the posterior distribution by the prior, and use the marginal log-likelihood to calculate the maximum likelihood estimate and *P*-value.

Other evidence supporting causality

For each putative core gene, we examined six other possible sources of orthogonal evidence for causality, as previously: (1) association of *cis*-eQTL variants (within 5 Mb of the transcription site) with MAP; (2) reported association with essential hypertension of SNPs within 200 kb of the transcription start site in a published meta-GWAS[23]; (3) effects of rare variants in the gene on blood pressure or related traits; (4) association of blood pressure with measured levels of the

transcript (for eQTL GATE associations) or the circulating protein (for pQTL GATE associations) in the same direction as the association with the GATE score; (5) reports that gene knockout, overexpression, or pharmacological perturbation alters blood pressure in experimental models; (6) reports that targeting the gene product or its ligand/receptor alter blood pressure.

For effects of rare variants on blood pressure, we queried two public databases of gene-based rare-variant association statistics: Genebass (app.genebass.org) and the AstraZeneca PheWAS portal (azphewas.com). These platforms report results from burden tests across a wide range of clinical outcomes in exome-sequenced UK Biobank participants. **add Canadian study https://www.sciencedirect.com/science/article/pii/S0828282X23015167**

Tests for association of proteins with mean arterial pressure were based on the UK Biobank Pharma Proteomics Project, a subset of 53014 participants in the main cohort. This resource provides individual-level data for 2,923 circulating proteins assayed on the Olink panel across 53,014 participants. We excluded proteins with greater than 20% missingness. For the remaining proteins, missing values were imputed via an Expectation-Maximization algorithm that estimates values based on the covariance structure of the observed data, as implemented in the Python package `ppca`. Association of transcript levels in whole blood (Gieger paper) …

Results

The procedure for examining eQTL or pQTL GATE score associations with a trait is quite unlike the workflow for a GWAS. First, we list the genes that are the target of multiple *trans*-QTLs and for which the GATE score is associated with the trait. We denote these genes as putative core genes, and examine a heatmap of correlations between the GATE scores for these genes. A block of highly correlated GATE scores indicates that the target genes share *trans*-QTLs with pleiotropic effects, so that a causal role of any one of these genes cannot reliably be inferred without other evidence. Second, we examine a bipartite graph of putative core genes and their *trans*-QTLs, where the *trans*-QTLs are restricted to those that contain reported GWAS hits and the target genes are restricted to those with some minimum number of edges from these GWAS hit-containing *trans*-QTLs. Core genes, by definition, are those on which the effects of multiple trait-associated common variants coalesce to influence the trait. Third, we examine the table of putative core genes to see the magnitude and direction of GATE score associations, and any supporting evidence from *cis*-QTL associations or reported GWAS hits. Fourth, we test for a dose-response effect as described above. Fifth, we examine other supporting evidence: effects of rare variants in the gene, association of the trait with measured levels of the transcript or protein, perturbation of the gene in experimental models, and drug effects in humans.

Association of MAP with GATE scores for gene expression

Figure 1, shows a heatmap of correlations between the 28 eQTL GATE scores that have effective number of *trans*-QTLs at least 8 and are associated with mean arterial pressure at $P < 10^{-5}$. The GATE scores are ordered by hierarchical clustering on the squared correlations so that highly correlated scores appear in diagonal blocks. This shows a cluster of highly correlated scores for genes that are expressed on B cells, including *BANK1*, *COBLL1* and *MEF2C*.

Figure 2 shows that these GATE scores are co-regulated by the *SH2B3* locus on chromosome 12, which appears to be a "peripheral master regulator". This Figure shows no clear evidence for core genes on which many GWAS hit genes coalesce via *trans* effects. This is likely to reflect the limitations of the eQTLGen summary statistics, in which most SNPs were not tested for *trans*-effects.

Table 1 shows that the GATE scores that form the cluster of B cell genes are inversely associated with MAP. The GATE score association with MAP was supported by a *cis*-eQTL association for three of the putative core genes, and supported by reported GWAS hit within 200 kb for 11 of the putative core genes, although most of these GWAS hits had not been attributed to a nearby gene in the original GWAS.

Association of mean arterial pressure with GATE scores for circulating proteins

Figure 3 shows a heatmap of correlations betweeen the 24 pQTL GATE scores that have effective number of *trans*-pQTLs at least 20 and are associated with mean arterial pressure at $P < 10^{-5}$. Only two GATE scores – for *CNTN3* and *CKB* – are highly correlated. For the other genes, it is less likely that GATE score associations with blood pressure can be explained by pleiotropic effects.

Figure 4 shows the putative core genes that are targets of at least 10 *trans*-pQTLs that contain GWAS hits. The coalescence of multiple trait-associated *trans*-pQTLs on these target genes, combined with the result that the aggregated *trans*-effect is associated with the trait, supports the designation of these genes as core genes for blood pressure. The most extreme gene is *CD300LG*, on which 18 GWAS hit-containing *trans*-pQTLs coalesce.

Table 2 shows that the most strongly associated pGATE scores was for *IL17RB*. Of the 24 putative core genes shown, 11 had a reported GWAS hit within 200kb most of which could not be localised to any gene in the originating GWAS.

A full listing of all eGATE and pGATE scores (N=210) that were associated with MAP at $P < 10^{-5}$ but where the effective number of loci threshold is relaxed to ≥ 5 is available as a Supplementary file.

Dose-response (Mendelian randomization) analysis

Supplementary Tables S2 and S3 show the results of the dose-response analysis for putative core genes listed in Table 1 and Table 2. Of the eQTL GATE scores for putative core genes only *ABCA1* showed a dose-response relationship at $P < 0.01$. As the statistical power to detect a dose-response relationship depends upon the number of trans-QTLs, the tests for eQTL GATE scores have relatively low statistical power. Of the pQTL GATE scores for putative core genes, 13 of 24 genes in Table 2 showed a dose-response effect on mean arterial pressure at $P < 0.01$). Figure S1 shows scatter plots of the dose-response relationship betwen the effects of each *trans*-pQTL on protein levels and the effects of each *trans*-pQTL on blood pressure, for the associations for the top 6 genes that show a dose-response relationship: *IL17RB*, *CEACAM16*, *PTPRF*, *CKB*, *TCL1A* and *CNTN3*.

Other criteria for causality

Supplementary Table S1 summarizes for each putative core gene the results of the prespecified criteria other than *trans*-QTL effects that support causality of the gene for MAP. In addition to 24 genes having known GWAS hits or *cis* associations, 13 having significant MR support we note that

1. For none of the putative core genes have rare variants been reported to cause monogenic hypertension: most of the genes that cause monogenic hypertension are in the renin-angiotensin aldosterone system.

2. For 17 of the core genes, measured levels of the proteins encoded by these genes in the UKB-PPP study showed associations with MAP at $P < 10^{-4}$ in Supplementary Table S1. The directions of these associations with MAP were consistent with the GATE analysis for 11 core genes (Supplementary Tables S4 and S5 and Supplementary Figure S2). Because the associations of mean arterial pressure with measured protein levels are cross-sectional rather than prospective, these association may be confounded by co-morbidity and reverse causation.

3. For 7 of the core genes, we identified supportive evidence for a causal relationship based on perturbation in a mouse model of hypertension. For 5 genes (*TP53INP2*[24], *IRAG1*[25], *RTN1*[26], *CD300LG*[27], and *IGFBP2*[28]) the directions of effect of the genetically predicted levels of gene expression (for eGATE scores) or protein levels (for pGATE scores) on MAP in the GATE analysis are the same as that expected from animal models. Higher predicted levels of *IRAG1*, *TP53INP2*, *CD300LG*, and *IGFBP2* associate with lower MAP (negative slope), consistent with their protective roles in animal hypertension models, while higher predicted levels of *RTN1* associate with higher MAP (positive slope), in line with its pro-hypertensive renal role. For 2 genes (*IL17RB*[29] and *CKB*[30]) their directions of effect in the GATE analysis are the opposite to that expected from animal models. *IL17RB* shows a negative GATE slope, yet *IL-17A* knock out blunts angiotensin II-induced hypertension in mice, implying higher *IL-17* signalling raises blood pressure; *CKB* similarly shows a negative GATE slope whereas CK inhibition lowers blood pressure in spontaneously hypertensive rats.

4. For one gene the proteins encoded by these genes or other proteins in the known pathway these proteins are involved in are in drug development for conditions that affect blood pressure: *TREM1*[31] (septic shock).

Discussion

Summary of principal findings

As the SNP effects on blood pressure are highly polygenic, so also the genes on which their *trans*-effects coalesce are more numerous, and have weaker effects than in our previous studies of autoimmune diseases that are relatively oligogenic. The largest effect that we identify is for *IL17B*, where an increase of 1 SD in the measured protein level lowers blood pressure by only 1.2 mmHg and an increase of 1 SD in the GATE score (which accounts for only XX% of the variance in the

measured protein level) lowers blood pressure by 0.1 mmHg. We summarize below the how the genes that we identify through aggregated *trans*-effects fall into relevant pathways: endothelial cell function, adaptive immune system, inflammatory response, and renin-angiotensin system.

Genes that act on endothelial cells

- Our results show that raised blood pressure is associated with genetically altered expression (as transcript levels in whole blood or circulating protein levels) of genes that act on or are expressed by endothelial cells. This includes higher expression of *CD36*, and lower levels of the proteins encoded by *CD300LG*, *ADIPOQ* and *DSG2*.
- *CD300LG* is expressed predominantly on vascular endothelial cells, and has recently been shown to act as a receptor for triglyceride-rich lipoproteins (chylomicrons and VLDL) that facilitates postprandial lipid clearance[32].
- *CD36* encodes the scavenger receptor B2 that mediates transport of nonesterified fatty acids across the endothelial cell barrier[33] through packaging with ceramides into exosome-like small extracellular vesicles that are released at the basolateral membrane[34]. CD36 deficiency causes resistance to insulin-mediated glucose uptake in hypertensive rats (Aitman TJ, et al. Identification of Cd36 (Fat) as an insulin-resistance gene causing defective fatty acid and glucose metabolism in hypertensive rats. Nat Genet. 1999;21:76–83. doi: 10.1038/5013) and in humans[35]. Both *cis*- and *trans*-effects on expression of *CD36* in whole blood are positively associated with blood pressure.
- *ADIPOQ* encodes adiponectin which is produced by adipocytes, circulates at micromolar concentrations, and is taken up by endothelial cells via CDH13 receptor where it forms exosomes containing ceramides. It appears to act as a garbage collector[36,37].
- *DSG2*, expressed on endothelial cells, encodes a receptor for the fatty acid binding protein FABP4[38].

A possible mechanism by which altered lipid handling in vascular endothelial cells might raise blood pressure is through accumulation of lipid droplets[39]. Endothelial lipid droplet accumulation in response to high-fat diet reduces nitric oxide production and raises blood pressure[40].

Genes that act on the adaptive immune system

- Our results show that elevated blood pressure is associated with genetically down-regulated expression in whole blood of of a cluster of co-regulated genes that are expressed on B cells, including *BANK1*, *COBLL1* and *TNFRSF13B*. This cluster of genes may represent a subtype of B cells. For *BANK1*, causality is supported by GWAS associations. The role of B cells in hypertension is supported by experiments showing that in which the hypertensive response to angiotensin II infusion is attenuated by genetic knockout or pharmacological depletion of B cells (Drummond, G., Bobik, A., Toh, B.-H., Kyaw, T., Guzik, T., Peter, K., Samuel, C., Vinh, A., Tipping, P., Salimova, E., Lewis, C., Krishnan, S., Kim, H., Diep, H., Kett, M., Ferens, D., Lieu, M., Sobey, C., & Chan, C. (2015). Obligatory Role for B Cells in the Development of Angiotensin II–Dependent Hypertension. Hypertension, 66(5), 1023–1033. https://doi.org/10.1161/HYPERTENSIONAHA.115.05779)

- Only a few of the genes implicated in hypertension by GATE analysis act on T cells or NK cells. *CKB* (encoding creatine kinase B) modulates thymocyte selection, T cell activation and T cell receptor-mediated signaling (Samborska B, Roy DG, Rahbani JF, Hussain MF, Ma EH, Jones RG, Kazak L. Creatine transport and creatine kinase activity is required for CD8+ T cell immunity. Cell Rep. 2022 Mar 1;38(9):110446. doi: 10.1016/j.celrep.2022.110446.). *MEF2C* expressed by NK cells, encodes a transcription factor that regulates NK cell function via effects on lipid metabolism. The most interesting gene related to T cell function is *BTN2A1* (encoding butyrophilin 2A1), which is supported by a dose-response relationship and by a reported GWAS hit. Butyrophilin 2A1 has a key role in the immune response to intracellular phosphoantigens, which are associated with bacterial infection or malignancy. Intracellular phosphoantigens glue together the intracellular domains of butryophilin 2A1 and butyrophilin 3A1 (encoded by *BTN3A1*). The extracellular domain of this heterodimer is recognized by Vγ9Vδ2 γδ T cells (Cano CE. BTN2A1, an immune checkpoint targeting Vγ9Vδ2 T cell cytotoxicity against malignant cells. Cell Rep. 2021 Jul 13;36(2):109359. doi: 10.1016/j.celrep.2021.109359). (Yuan, L., Ma, X., Yang, Y. et al. Phosphoantigens glue butyrophilin 3A1 and 2A1 to activate Vγ9Vδ2 T cells. Nature 621, 840–848 (2023). https://doi.org/10.1038/s41586-023-06525-3)

Rare loss of function variants in *BTN3A1* are associated with increased blood pressure. https://www.sciencedirect.com/science/article/pii/S0828282X23015167

The role of γδ T cells in hypertension is supported by experimental models in which genetic knockout or pharmacological depletion of γδ T cells attenuates the hypertensive response to angiotensin II infusion. https://www.ahajournals.org/doi/10.1161/circulationaha.116.027058. Depletion of Vγ6Vδ1 T cells however increases the hypertensive response to angiotensin II, suggesting that specific subtypes of γδ T cells have different effects on blood pressure. (Mahmoud AUM, Caillon A, Shokoples B, Ferreira NS, Comeau K, Hatano S, Yoshikai Y, Lewis JM, Tigelaar RE, Paradis P, Schiffrin EL. Vγ6/Vδ1 + γδ T cells protect from angiotensin II effects on blood pressure and endothelial function in mice. J Hypertens. 2025 Jan 1;43(1):109-119. doi: 10.1097/HJH.0000000000003871)

Genes that act on inflammatory responses.

The GATE score for *IL17RB* is inversely associated with blood pressure, and the GATE score for *IL2RB* is positively associated with blood pressure. *IL17RB* encodes interleukin-17 receptor B which forms a heterodimer with interleukin-17 receptor A that is the binds interleukin-25 (alias IL-17E). Interleukin-25 signalling via this pathway induces the expression of *CXCL8* (alias IL-8). No experimental studies of effects of *IL17RB* perturbation on blood pressure have been reported, though an interleukin-17 receptor B inhibitor has reached clinical stage for atopic dermatitis[41].

IL2RB forms two heterotrimers, one of which is the receptor for interleukin-2, and the other for interleukin-15. Low-dose infusion of interleukin-2 normalized hypertension in a rat model of placental ischemia. (Deer E, Amaral LM, Campbell N, Fitzgerald S, Herrock O, Ibrahim T, LaMarca B. Low Dose of IL-2 Normalizes Hypertension and Mitochondrial Function in the RUPP Rat Model of Placental Ischemia. Cells. 2021 Oct 19;10(10):2797. doi: 10.3390/cells10102797)

- *TREM1* encodes a receptor belonging to the Ig superfamily that is expressed on myeloid cells. This protein amplifies neutrophil and monocyte-mediated inflammatory responses

- *CLEC4C* encodes a receptor for asialo-galactosyl-oligosaccharides that inhibits interferon production by plasmacytoid dendritic cells.
- *CD209* (alias dendritic cell-specific intercellular adhesion molecule-3–grabbing nonintegrin, DC-SIGN), is a C-type lectin receptor that is expressed primarily on subsets of dendritic cells and certain macrophages. It recognizes and binds to specific carbohydrate moieties—such as high-mannose structures and fucose-containing glycans expressed by pathogens.

Genes that act on the renin-angiotensin system or the kidney

Pro-inflammatory cytokines such as IL6, TNF-α, IL1β, IL18 and CCL2 predict incident hypertension[8,42].

The only gene identified through GATE analysis that has effects on the renin-angiotensin system is *CPA1*, encoding carboxypeptidase 1 which degrades angiotensin II[43]. The only gene that is known to affect the kidney is *RTN1* which encodes reticulon-1. Overexpression in renal cells induces ER stress and apoptosis. Knockdown of *Rtn1a* expression *in vivo* attenuates ER stress, proteinuria, glomerular hypertrophy and mesangial expansion in diabetic mice.

Limitations

The most serious limitation of this study is the inadequacy of the eQTLGen Phase 1 dataset, in which only 10,316 trait-associated SNPs were tested for *trans*-effects on transcript levels. This limitation will be overcome when summary statistics from the eQTLGen Phase 2 study are released. Although the eQTLGen summary statistics are not directly adjusted for cell type proportions, they were adjusted for principal components which contain information about cell type proportions. Pleiotropy from direct *trans*-QTL effects on disease may confound the inference of core genes, where a gene may appear associated with disease solely because its GATE score contains *trans*-QTLs from regions with large direct effects. For this reason the key results are restricted to genes that have a minimum effective number of _trans-QTLs, in which the relative contribution of a few peripheral master regulators is less.

For GATE analysis of *trans*-effects on circulating proteins, coverage is limited to the proteins on the SomaScan version 4 and Olink Explore 3072 platforms. Coverage of proteins will improve when the next tranche of UK Biobank proteomics data is released. Although GATE analysis is limited to *trans*-effects on transcripts in whole blood and proteins in plasma, many tissue-specific proteins are present at measurable concentrations in plasma.

GATE analysis cannot alone establish the direction of effect of a core gene. For instance where a gene encodes a cellular receptor such as *IL17RB*, genetic effects that increase levels of the circulating (soluble) form of the protein may have effects opposite in direction to the effects of signaling through the cellular receptor.

For a continuous trait such as blood pressure, GATE analysis based on published summary statistics for transcript or protein levels cannot exclude reverse causation. This would require recomputing summary statistics in normotensive individuals, or in a younger cohort in whom hypertension is rare.

Perspectives

These results identify several pathways that may be amenable to the development of new pharmacological interventions. Specifically, they point to genes that regulate lipid handling in endothelial cells, including *CD300LG*, *CD36*, and *ADIPOQ*. This is consistent with experimental evidence that lipid accumulation in endothelial cells impairs endothelial function and raises blood pressure. The results also broadly support a role for the adaptive immune system and for interleukin-dependent inflammatory responses in hypertension. Specifically they point to the *BTN2A1*/*BTN3A1*-mediated recognition of intracellular phosphoantigens by $\gamma\delta$ T cells.

Novelty and relevance

What is new?

What is relevant?

Clinical/pathophysiological implications

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Disclosures

The authors declare no conflicts of interest relevant to this article.

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Tables

Table 1. Putative core genes associated with mean arterial pressure at $P < 10^{-5}$, of 235 with effective number of *trans*-eQTLs > 8. Rows ordered as in Fig 1.

	Transcription site		*trans* score			*cis* score		
Gene	Chr	Start position (Mb)	Effective number of *trans*-eQTLs	Slope	P value	Slope	P value	Reported GWAS hit
ABCA1	9	104.78	10.0	-0.009	2×10^{-6}	0.001	0.7	.
ALOX5AP	13	30.71	8.1	-0.009	8×10^{-7}	0.000	0.8	.
CCN3	8	119.42	13.8	-0.010	2×10^{-7}	-0.002	0.4	.
KCNJ15	21	38.16	8.2	0.011	1×10^{-8}	-0.002	0.2	.
GAS7	17	9.91	13.9	-0.008	6×10^{-6}	0.001	0.6	.
MS4A6A	11	60.17	9.5	-0.009	4×10^{-7}	0.002	0.4	***MS4A6A***
DPYSL2	8	26.51	8.8	-0.010	1×10^{-7}	-0.009	3×10^{-6}	.
SLC31A2	9	113.15	9.4	-0.013	6×10^{-13}	0.000	0.9	.
TREM1	6	41.27	8.1	-0.019	5×10^{-25}	-0.003	0.2	+
TP53INP2	20	34.70	8.1	-0.012	6×10^{-10}	0.002	0.2	.
IL2RB	22	37.12	9.8	0.012	1×10^{-10}	-0.003	0.08	+
COLQ	3	15.45	8.9	0.010	2×10^{-8}	0.002	0.4	+
IRAG1	11	10.57	8.5	-0.011	8×10^{-10}	-0.002	0.4	+
DHRS9	2	169.06	8.6	0.012	3×10^{-11}	0.002	0.3	+
TNFRSF13B	17	16.93	14.6	-0.011	1×10^{-8}	0.002	0.3	.
RRP12	10	97.36	8.3	-0.010	3×10^{-7}	0.000	0.8	.
CXCR5	11	118.88	10.5	-0.009	2×10^{-6}	-0.003	0.1	.
COBLL1	2	164.65	9.7	-0.010	2×10^{-7}	-0.005	0.02	.
BANK1	4	101.41	9.9	-0.012	3×10^{-11}	0.007	3×10^{-4}	***BANK1***
MEF2C	5	88.72	8.5	-0.009	4×10^{-6}	0.001	0.8	***LINC00461***
BCL11A	2	60.45	9.9	-0.009	2×10^{-6}	0.001	0.7	.
TESC	12	117.04	9.9	0.009	4×10^{-6}	0.002	0.3	.
UBB	17	16.38	15.1	0.009	1×10^{-6}	.	.	.
CD36	7	80.37	9.3	0.008	5×10^{-6}	0.004	0.02	+
RTN1	14	59.60	9.3	0.010	2×10^{-7}	-0.002	0.2	.
SFTPD	10	79.94	9.2	-0.015	6×10^{-16}	0.001	0.7	+
EMP1	12	13.20	10.2	-0.011	2×10^{-8}	0.001	0.5	+
GPBAR1	2	218.26	10.4	-0.009	3×10^{-6}	0.007	3×10^{-4}	.

Chr, chromosome; Mb, megabase; Slope, regression coefficient representing the change in MAP (mmHg) per SD increase in the eGATE score. For the analysis 212,569 individuals were included. Reported GWAS hits for essential hypertension are those falling within 200 kb of the transcription start site; (+) indicates that the SNP has not been attributed to a specific gene.

Table 2. Putative core genes associated with mean arterial pressure at $P < 10^{-5}$, of 280 with effective number of *trans*-pQTLs > 20. Rows ordered as in Fig 3.

		Transcription site		*trans* score			*cis* score		
pQTL study	Gene	Chr	Start position (Mb)	Effective number of *trans*-pQTLs	Slope	P value	Slope	P value	Reported GWAS hit
UKB	*APOD*	3	195.57	20.5	-0.012	8×10^{-10}	-0.003	0.2	**.**
UKB	*ADIPOQ*	3	186.84	20.7	-0.015	3×10^{-13}	0.000	1	**+**
UKB	*BTN2A1*	6	26.46	22.2	0.009	3×10^{-6}	.	.	***HFE***
UKB	*CD300LG*	17	43.85	23.6	-0.014	4×10^{-12}	-0.001	0.8	**.**
UKB	*DSG2*	18	31.50	20.2	-0.012	1×10^{-9}	0.000	0.9	**.**
UKB	*CNTN3*	3	74.26	33.5	0.013	3×10^{-11}	-0.001	0.5	**.**
UKB	*CKB*	14	103.52	20.7	-0.014	1×10^{-11}	.	.	**+**
UKB	*IGFBP2*	2	216.63	21.7	-0.013	3×10^{-11}	-0.001	0.5	**.**
DeC	*ELANE*	19	0.85	21.3	0.009	8×10^{-7}	0.001	0.6	**.**
DeC	*COCH*	14	30.87	20.7	-0.008	9×10^{-6}	0.002	0.3	**.**
UKB	*GCNT1*	9	76.42	22.3	-0.011	2×10^{-8}	0.002	0.2	**.**
UKB	*CD209*	19	7.74	25.9	0.009	6×10^{-6}	0.001	0.8	**+**
UKB	*CPM*	12	68.84	21.5	0.011	5×10^{-8}	0.000	1	**.**
DeC	*REG3G*	2	79.03	24.7	-0.008	1×10^{-5}	0.001	0.6	**+**
DeC	*CTRB1*	16	75.22	20.2	-0.010	4×10^{-8}	0.005	0.03	***LOC105371345***
UKB	*CPA1*	7	130.38	29.2	-0.010	4×10^{-7}	0.000	0.8	**+**
UKB	*CLEC4C*	12	7.73	24.8	0.010	2×10^{-7}	0.001	0.5	**+**
UKB	*TCL1A*	14	95.71	35.7	0.010	3×10^{-7}	0.004	0.07	**.**
UKB	*IL17RB*	3	53.85	23.6	-0.016	3×10^{-17}	-0.004	0.03	***CACNA1D***
UKB	*CEACAM16*	19	44.70	24.0	-0.013	2×10^{-11}	.	.	***APOE*, *PVRL2*, *TOMM40*, *APOC1*, *BCAM***
UKB	*VASN*	16	4.37	27.0	-0.012	2×10^{-9}	0.001	0.5	**.**
UKB	*PTPRF*	1	43.53	25.6	-0.014	3×10^{-12}	0.004	0.06	**+**
UKB	*MYOM3*	1	24.06	24.6	-0.010	8×10^{-7}	0.001	0.6	**.**
UKB	*EDA2R*	X	66.60	21.2	-0.010	9×10^{-7}	.	.	**.**

DeC, deCODE study; UKB, UK Biobank Pharma Proteomics Project; Chr, chromosome; Mb, megabase; Slope, regression coefficient representing the change in MAP (mmHg) per SD increase in the pGATE score. For DeCODE pQTLs the analysis is based on 212,569 individuals in UKBB. For UKB pQTLs the analysis is based on 190,635 individuals as individuals who were in the proteomics sub-study excluded. Reported GWAS hits for essential hypertension are SNPs within 200 kb of the transcription start site; (+) indicates that the SNP has not been attributed to a specific gene. Where multiple pGATE scores exist for a given protein, only the score with the strongest association ($P < 10^{-5}$) is reported.

Figure Legends

Fig 1. Heatmap of pairwise correlations between eQTL GATE scores in Table 1. Rows and columns are ordered by hierarchical clustering on the squared correlation coefficients.

Fig 2. Bipartite graph of *trans*-eQTL clumps containing GWAS hits for blood pressure or hypertension and their target genes from Table 1. Square nodes (top) represent *trans*-eQTL clumps labelled by chromosome and reported GWAS gene(s); circular nodes (bottom) represent target core genes sized by $-\log_{10}(P)$ for association of the eGATE score with MAP and coloured by direction of effect. Only genes connected to 2 or more GWAS-hit clumps are shown.

Fig 3. Heatmap of pairwise correlations between pQTL GATE scores in Table 2. Rows and columns are ordered by hierarchical clustering on the squared correlation coefficients.

Fig 4. Bipartite graph of *trans*-pQTL clumps containing GWAS hits for blood pressure or hypertension and their target genes from Table 2. Square nodes (top) represent *trans*-pQTL clumps labelled by chromosome and reported GWAS gene(s); circular nodes (bottom) represent target core genes sized by $-\log_{10}(P)$ for association of the pGATE score with MAP and coloured by direction of effect. Only genes connected to 10 or more GWAS-hit clumps are shown.

Figures

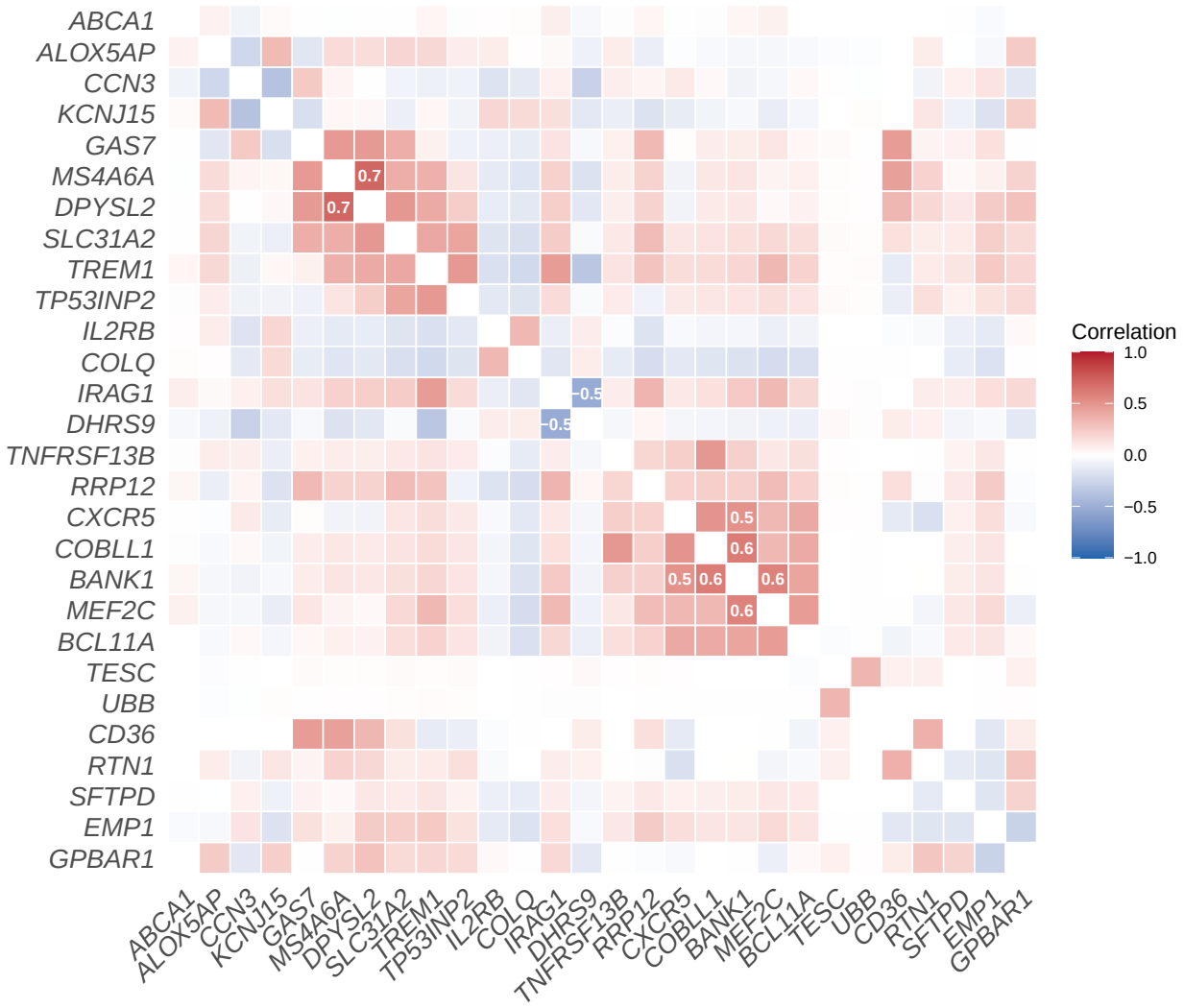

Fig 1. Heatmap of pairwise correlations between eQTL GATE scores in Table 1. Rows and columns are ordered by hierarchical clustering on the squared correlation coefficients.

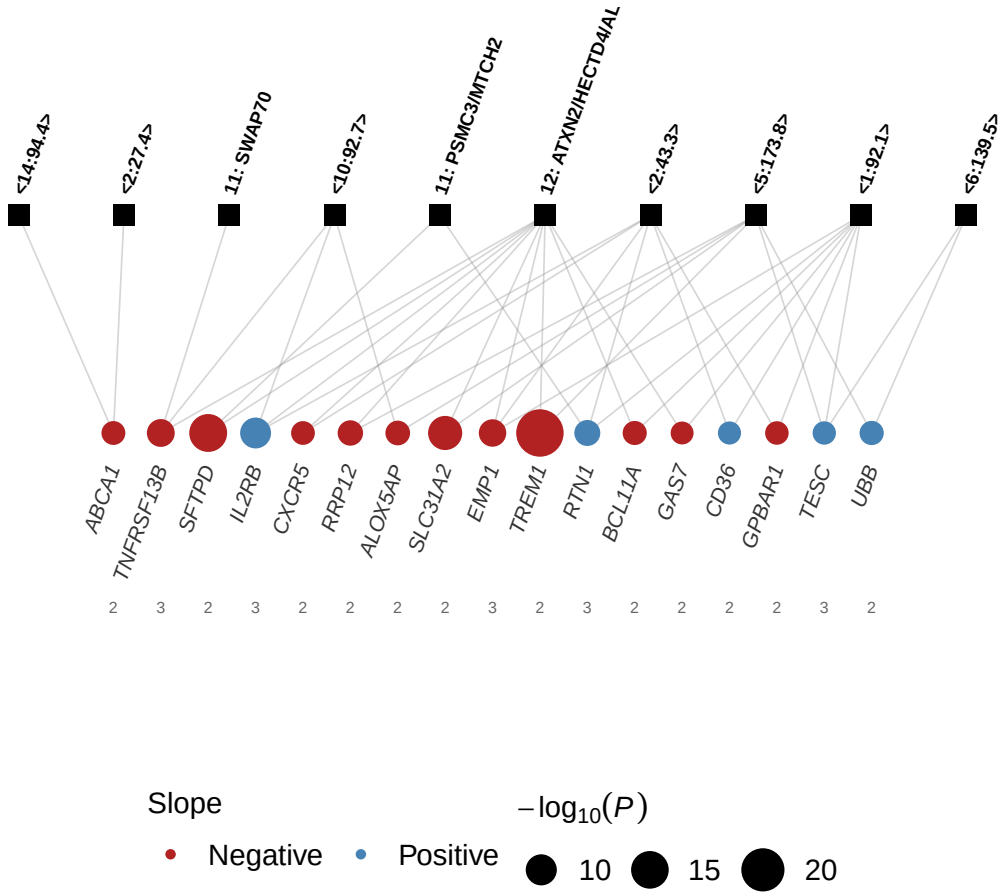

Fig 2. Bipartite graph of *trans*-eQTL clumps containing GWAS hits for blood pressure or hypertension and their target genes from Table~1. Square nodes (top) represent *trans*-eQTL clumps labelled by chromosome and reported GWAS gene(s); circular nodes (bottom) represent target core genes sized by $-\log_{10}(P)$ for association of the eGATE score with MAP and coloured by direction of effect. Only genes connected to 2 or more GWAS-hit clumps are shown.

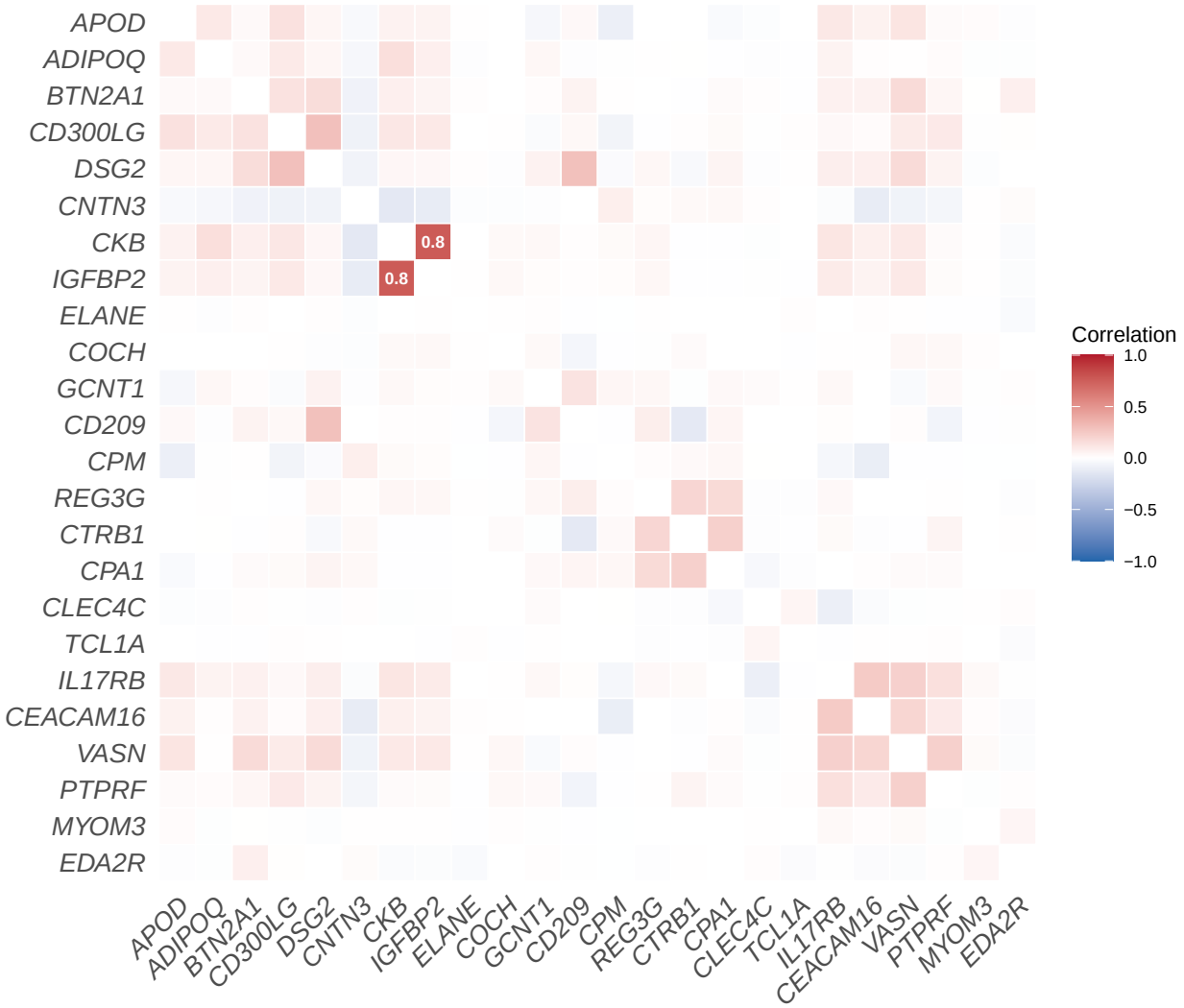

Fig 3. Heatmap of pairwise correlations between pQTL GATE scores in Table 2. Rows and columns are ordered by hierarchical clustering on the squared correlation coefficients.

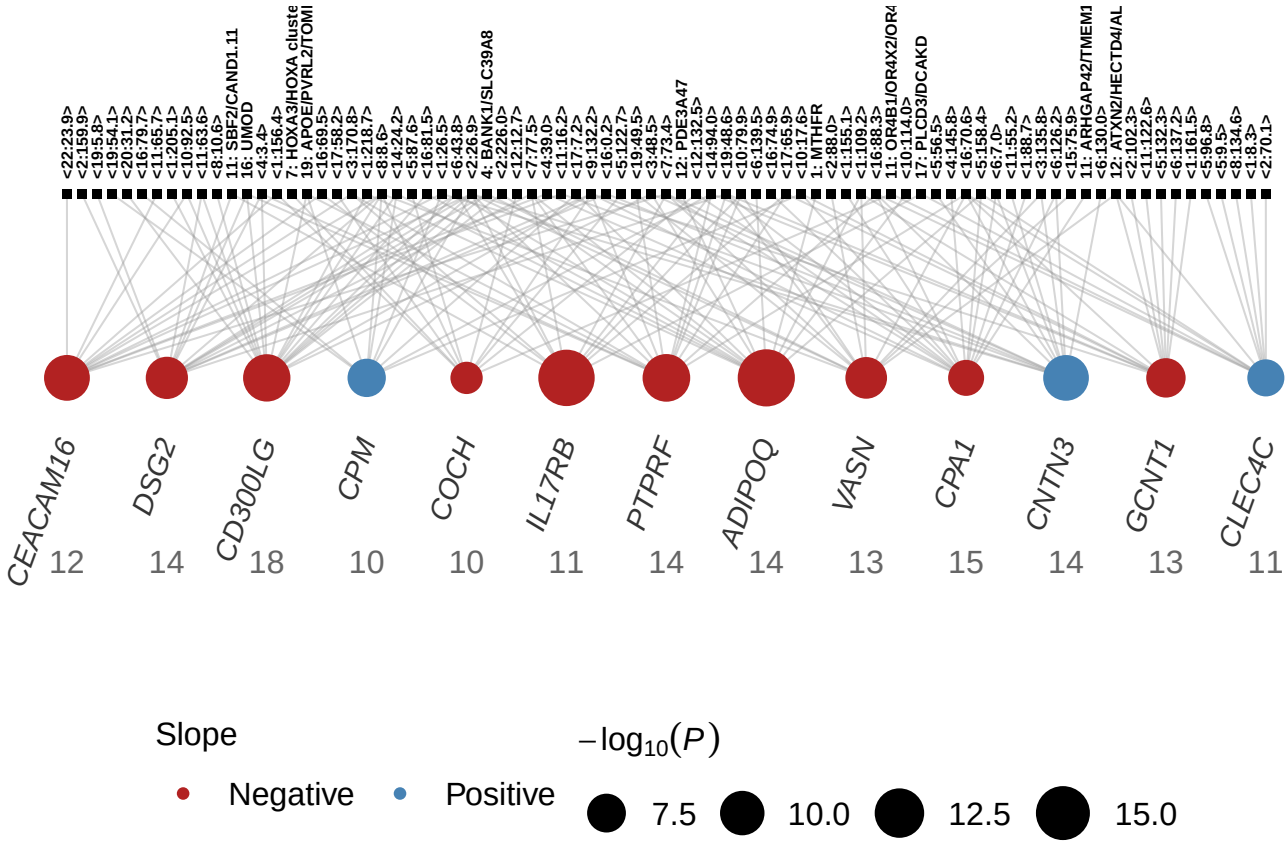

Fig 4. Bipartite graph of *trans*-pQTL clumps containing GWAS hits for blood pressure or hypertension and their target genes from Table 2. Square nodes (top) represent *trans*-pQTL clumps labelled by chromosome and reported GWAS gene(s); circular nodes (bottom) represent target core genes sized by $-\log_{10}(P)$ for association of the pGATE score with MAP and coloured by direction of effect. Only genes connected to 10 or more GWAS-hit clumps are shown.

Supplementary information

Supplementary Figures

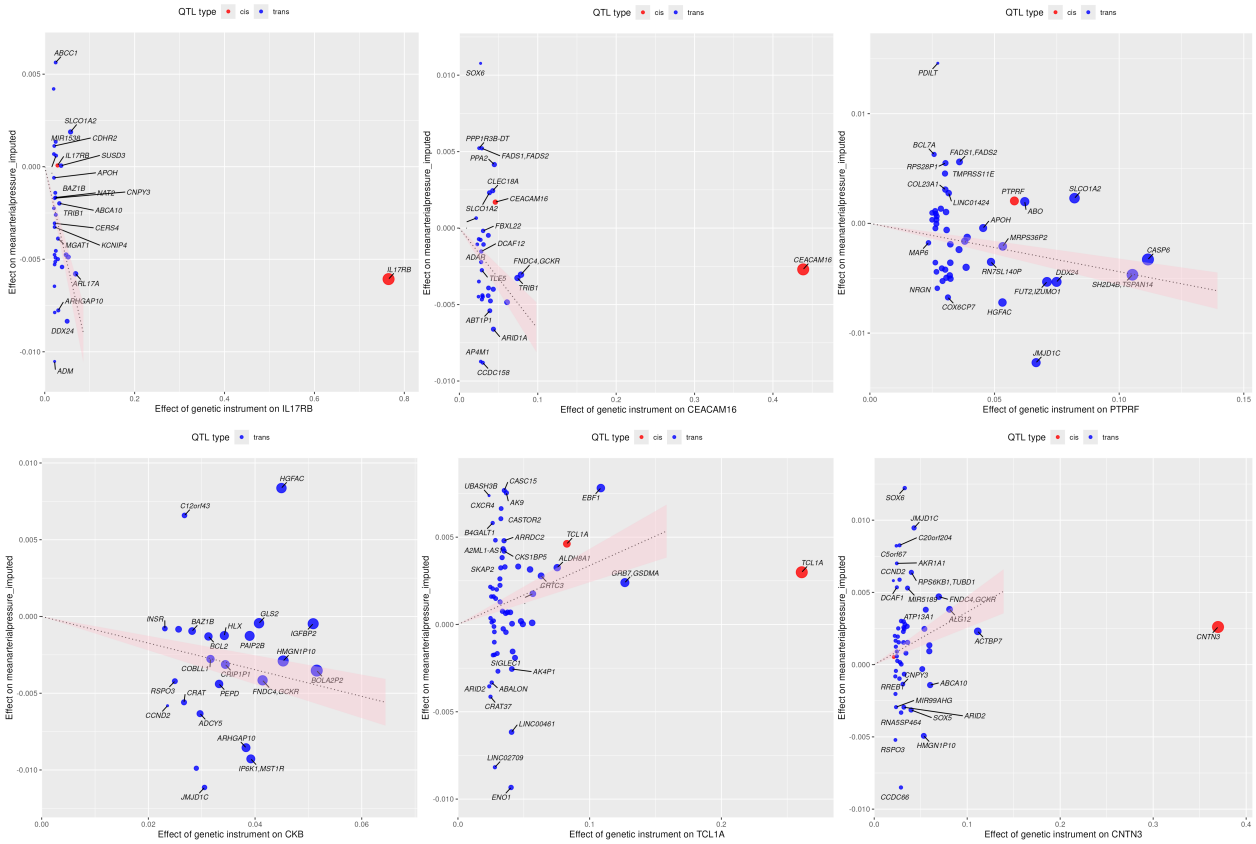

Fig S1. Mendelian randomisation analysis of the effect of the putative core proteins on MAP in UKB-PPP. All trans-pQTLs contributing to the pGATE score of the putative core proteins are used as genetic instrumens. The effects of the genetic instrument on protein level and MAP are shown on the x-axis and y-axis, respectively. The size of each point is inversely proportional to the standard error of the ratio estimator. The slope of the straight line represents the maximum likelihood estimate of the causal effect parameter while the shaded ribbon reflects one standard error above and below this maximum likelihood estimate.

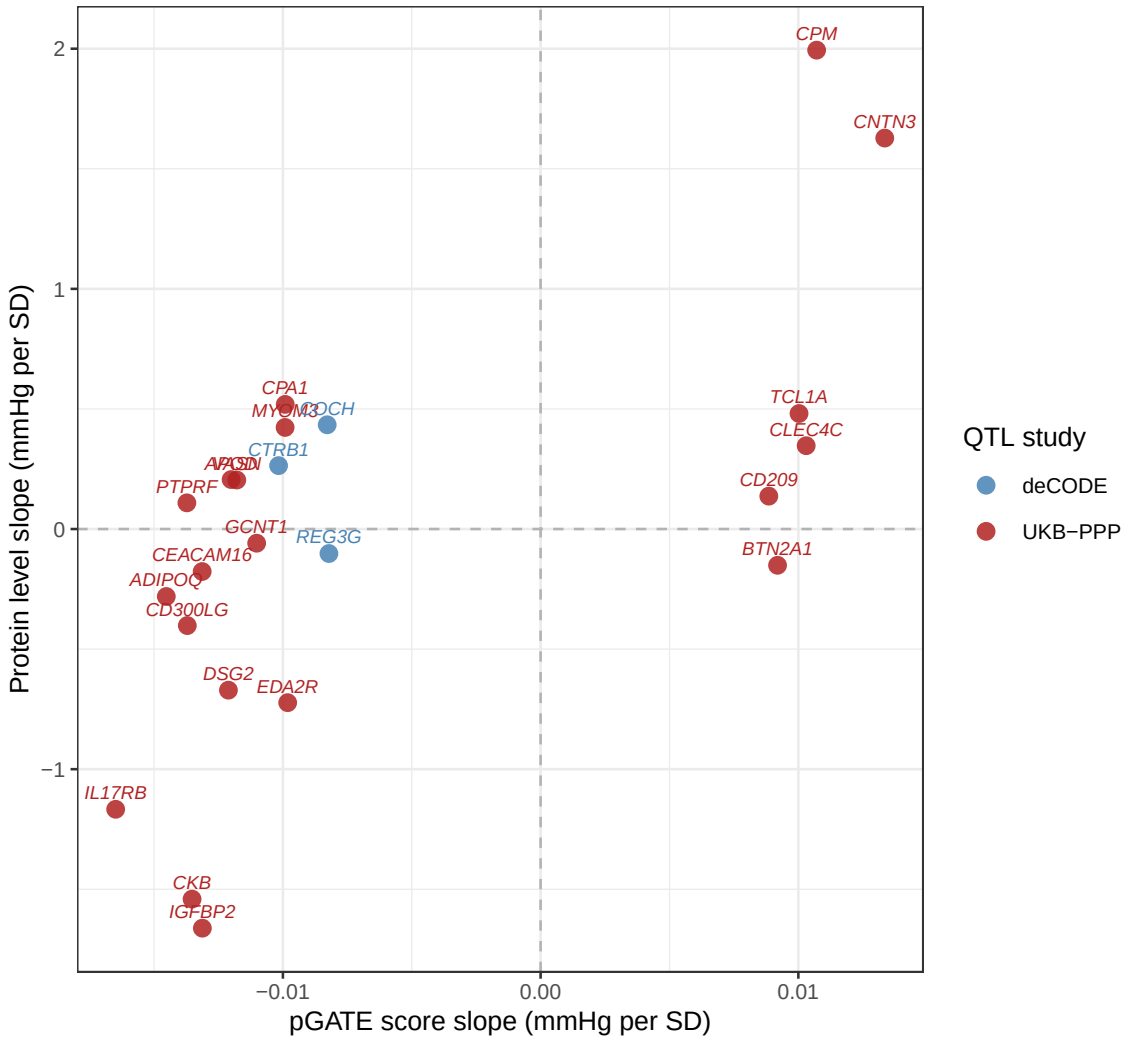

Fig S2. Scatter plot comparing the regression coefficient for the association of MAP with the pGATE score (x-axis) against the regression coefficient for the association of MAP with the measured protein level (y-axis) for putative core genes in Table 2. Colours indicate the pQTL study used to derive the pGATE score (deCODE or UKB-PPP). Gene symbols label each point. Only genes for which proteins weere measured in UKBB-PPP are shown.

Supplementary Tables

Table S1. Additional evidence in support of GATE-detected core genes for MAP with thresholds $P < 10^5$, effective numbers of eQTL ≥ 8 and pQTLs ≥ 20.

Gene symbol	GWAS hit[a]	Dose-response[b]	Monogenic cause[c]	Measured gene product[d]	Experimental validation[e]	Drug effect[f]
Analysis based on transcripts						
TREM1	+	.	−	.	−	+
SFTPD	+	.	−	0	−	−
SLC31A2	−	.	−	.	−	−
DHRS9	+	.	−	.	−	−
BANK1	+	.	−	−	−	−
IL2RB	+	.	−	0	−	−
TP53INP2	−	.	−	.	+	−
IRAG1	+	.	−	.	+	−
TNFRSF13B	−	.	−	0	−	−
KCNJ15	−	.	−	.	−	−
EMP1	+	.	−	.	−	−
COLQ	+	.	−	.	−	−
DPYSL2	+	.	−	.	−	−
RTN1	−	.	−	.	+	−
COBLL1	−	.	−	.	−	−
CCN3	−	.	−	−	−	−
RRP12	−	.	−	.	−	−
MS4A6A	+	.	−	.	−	−
ALOX5AP	−	.	−	.	−	−
UBB	−	.	−	.	−	−
BCL11A	−	.	−	.	−	−
ABCA1	−	.	+	.	−	−
CXCR5	−	.	−	.	−	−
GPBAR1	+	.	−	.	−	−
TESC	−	.	−	.	−	−
MEF2C	+	.	−	.	−	−
CD36	+	.	−	+	−	−
GAS7	−	.	−	.	−	−
Analysis based on circulating proteins						
IL17RB	+	+	−	+	+	−
ADIPOQ	+	−	+	0	−	−
PTPRF	+	+	−	0	−	−
CD300LG	−	−	−	+	+	−
CKB	+	+	−	+	+	−
CEACAM16	+	+	−	0	−	−
CNTN3	−	+	−	+	−	−
IGFBP2	−	+	−	+	+	−
APOD	−	+	−	0	−	−
DSG2	−	−	+	+	−	−
VASN	−	−	−	0	−	−
GCNT1	−	+	−	0	−	−
CTRB1	+	−	−	−	−	−
CPM	−	+	−	+	−	−
CLEC4C	+	−	−	+	−	−
TCL1A	−	+	−	+	−	−
CPA1	+	−	−	−	−	−
MYOM3	−	−	−	−	−	−
ELANE	−	−	+	.	−	−
EDA2R	−	+	+	+	−	−
BTN2A1	+	+	−	0	−	−
CD209	+	−	−	0	−	−
COCH	−	+	−	−	−	−
REG3G	+	−	−	0	−	−

Symbols: + indicates that the criterion is met; −− indicates that it is not; . data is missing.
(a) Association with essential hypertension reported for SNP within 200 kb, or *cis*-score for the gene is associated with MAP ($P < 0.001$).
(b) Mendelian randomization analysis based on *trans*-QTLs supports a causal effect at $P < 0.01$ and effective number of eQTLs/pQTLs ≥ 20.
(c) Rare variants in the gene cause monogenic form of the blood pressure-related diseases.
(d) Association of MAP with measured protein levels in UKB-PPP: + indicates association at $P < 10^{-4}$ with effect direction consistent with GATE score effect on MAP; − indicates opposite direction; 0 indicates no association.
(e) Perturbation of the gene in an experimental model alters blood pressure
(f) Drug that targets the protein encoded by the gene or its ligand/receptor alters blood pressure in humans.

Table S2. Mendelian randomization analysis based on transcripts: dose-response relationship between *trans*-eQTL effects on gene expression and effects on MAP, for putative core genes in Table 1.

Gene	QTL study	Number of *trans*-QTLs	Estimate of causal effect parameter	P value
ABCA1	eQTLGen	11	-0.095	0.009
CD36	eQTLGen	10	0.069	0.01
GPBAR1	eQTLGen	12	-0.061	0.02
SFTPD	eQTLGen	11	-0.059	0.02
UBB	eQTLGen	18	0.048	0.03
TESC	eQTLGen	11	0.046	0.04
ALOX5AP	eQTLGen	10	-0.091	0.05
TNFRSF13B	eQTLGen	15	-0.071	0.09
SLC31A2	eQTLGen	11	-0.046	0.3
COLQ	eQTLGen	10	-0.027	0.3
RTN1	eQTLGen	13	0.019	0.3
IL2RB	eQTLGen	11	0.035	0.4
DHRS9	eQTLGen	12	0.029	0.5
BCL11A	eQTLGen	11	0.028	0.6
IRAG1	eQTLGen	11	-0.019	0.6
GAS7	eQTLGen	16	-0.014	0.6
EMP1	eQTLGen	12	-0.016	0.6
CCN3	eQTLGen	17	-0.006	0.8
BANK1	eQTLGen	11	-0.010	0.8
COBLL1	eQTLGen	10	0.008	0.8
CXCR5	eQTLGen	11	0.008	0.8
MS4A6A	eQTLGen	12	0.006	0.8
DPYSL2	eQTLGen	10	0.001	1

Genetic study of gene expression or protein levels in which instruments were created is shown in QTL study column. Only genes with at least 10 *trans*-QTLs available as instruments are included in this table.

Table S3. Mendelian randomization analysis based on circulating proteins: dose-response relationship between *trans*-pQTL effects on protein levels and effects on MAP, for putative core genes in Table 2.

Gene	QTL study	Number of *trans*-QTLs	Estimate of causal effect parameter	P value
IL17RB	UKBPPP	33	-0.105	1×10^{-8}
CEACAM16	UKBPPP	32	-0.066	7×10^{-5}
PTPRF	UKBPPP	46	-0.044	2×10^{-4}
CKB	UKBPPP	24	-0.087	2×10^{-4}
TCL1A	UKBPPP	58	0.034	4×10^{-4}
CNTN3	UKBPPP	53	0.036	8×10^{-4}
GCNT1	UKBPPP	29	-0.066	0.001
APOD	UKBPPP	28	-0.054	0.001
COCH	deCODE	39	-0.026	0.002
CPM	UKBPPP	35	0.052	0.002
IGFBP2	UKBPPP	23	-0.092	0.003
EDA2R	UKBPPP	24	-0.103	0.004
BTN2A1	UKBPPP	31	0.050	0.007
MYOM3	UKBPPP	41	-0.040	0.01
CLEC4C	UKBPPP	59	0.023	0.02
CD300LG	UKBPPP	40	-0.041	0.04
CPA1	UKBPPP	55	-0.020	0.04
ADIPOQ	UKBPPP	44	-0.038	0.04
ELANE	deCODE	24	0.034	0.09
REG3G	deCODE	28	-0.026	0.2
CTRB1	deCODE	25	-0.029	0.3
VASN	UKBPPP	33	-0.015	0.6
DSG2	UKBPPP	35	-0.008	0.6
CD209	UKBPPP	33	-0.004	0.6

Genetic study of gene expression or protein levels in which instruments were created is shown in QTL study column. Only genes with at least 10 *trans*-QTLs available as instruments are included in this table.

Table S4. Association of MAP with measured protein levels in the UKB-PPP study for the putative core genes identified by analysis based on transcripts (Table 1).

		protein association		*trans* score	
Gene	QTL study	Slope	P value	Slope	P value
TREM1	eQTLGen	.	.	-0.019	5×10^{-25}
SFTPD	eQTLGen	-0.117	0.07	-0.015	6×10^{-16}
SLC31A2	eQTLGen	.	.	-0.013	6×10^{-13}
DHRS9	eQTLGen	.	.	0.012	3×10^{-11}
BANK1	eQTLGen	0.341	9×10^{-8}	-0.012	3×10^{-11}
IL2RB	eQTLGen	0.015	0.8	0.012	1×10^{-10}
TP53INP2	eQTLGen	.	.	-0.012	6×10^{-10}
IRAG1	eQTLGen	.	.	-0.011	8×10^{-10}
TNFRSF13B	eQTLGen	0.142	0.03	-0.011	1×10^{-8}
KCNJ15	eQTLGen	.	.	0.011	1×10^{-8}
EMP1	eQTLGen	.	.	-0.011	2×10^{-8}
COLQ	eQTLGen	.	.	0.010	2×10^{-8}
DPYSL2	eQTLGen	.	.	-0.010	1×10^{-7}
RTN1	eQTLGen	.	.	0.010	2×10^{-7}
COBLL1	eQTLGen	.	.	-0.010	2×10^{-7}
CCN3	eQTLGen	0.705	2×10^{-23}	-0.010	2×10^{-7}
RRP12	eQTLGen	.	.	-0.010	3×10^{-7}
MS4A6A	eQTLGen	.	.	-0.009	4×10^{-7}
ALOX5AP	eQTLGen	.	.	-0.009	8×10^{-7}
UBB	eQTLGen	.	.	0.009	1×10^{-6}
BCL11A	eQTLGen	.	.	-0.009	2×10^{-6}
ABCA1	eQTLGen	.	.	-0.009	2×10^{-6}
CXCR5	eQTLGen	.	.	-0.009	2×10^{-6}
GPBAR1	eQTLGen	.	.	-0.009	3×10^{-6}
TESC	eQTLGen	.	.	0.009	4×10^{-6}
MEF2C	eQTLGen	.	.	-0.009	4×10^{-6}
CD36	eQTLGen	0.491	3×10^{-14}	0.008	5×10^{-6}
GAS7	eQTLGen	.	.	-0.008	6×10^{-6}

Table S5. Association of MAP with measured protein levels in the UKB-PPP study for the putative core genes identified by analysis based on circulating proteins (Table 2).

		protein association		*trans* score	
Gene	QTL study	Slope	P value	Slope	P value
IL17RB	UKB	-1.167	3×10^{-68}	-0.016	3×10^{-17}
ADIPOQ	UKB	-0.281	3×10^{-4}	-0.015	3×10^{-13}
PTPRF	UKB	0.109	0.1	-0.014	3×10^{-12}
CD300LG	UKB	-0.402	1×10^{-9}	-0.014	4×10^{-12}
CKB	UKB	-1.541	6×10^{-97}	-0.014	1×10^{-11}
CEACAM16	UKB	-0.177	0.01	-0.013	2×10^{-11}
CNTN3	UKB	1.628	4×10^{-115}	0.013	3×10^{-11}
IGFBP2	UKB	-1.662	2×10^{-110}	-0.013	3×10^{-11}
APOD	UKB	0.206	0.002	-0.012	8×10^{-10}
DSG2	UKB	-0.671	1×10^{-23}	-0.012	1×10^{-9}
VASN	UKB	0.204	0.002	-0.012	2×10^{-9}
GCNT1	UKB	-0.059	0.4	-0.011	2×10^{-8}
CTRB1	DeC	0.264	3×10^{-5}	-0.010	4×10^{-8}
CPM	UKB	1.994	2×10^{-164}	0.011	5×10^{-8}
CLEC4C	UKB	0.347	6×10^{-8}	0.010	2×10^{-7}
TCL1A	UKB	0.481	1×10^{-13}	0.010	3×10^{-7}
CPA1	UKB	0.519	8×10^{-16}	-0.010	4×10^{-7}
MYOM3	UKB	0.423	8×10^{-10}	-0.010	8×10^{-7}
ELANE	DeC	.	.	0.009	8×10^{-7}
EDA2R	UKB	-0.723	3×10^{-19}	-0.010	9×10^{-7}
BTN2A1	UKB	-0.151	0.03	0.009	3×10^{-6}
CD209	UKB	0.137	0.03	0.009	6×10^{-6}
COCH	DeC	0.434	4×10^{-11}	-0.008	9×10^{-6}
REG3G	DeC	-0.102	0.1	-0.008	1×10^{-5}

Table S6. Additional putative core genes defined by an association $P < 10^{-5}$ and effective *trans*-eQTLs contributing to the eGATE score not less than five but less than eight.

Gene	Transcription site		*trans* score			*cis* score		Reported GWAS hit
	Chr	Start position (Mb)	Effective number of *trans*-eQTLs	Slope	P value	Slope	P value	
SWAP70	11	9.66	6.0	-0.023	1×10^{-33}	-0.016	7×10^{-18}	***SWAP70***
TMEM204	16	1.53	6.8	0.023	2×10^{-32}	0.000	0.9	.
RUBCNL	13	46.33	5.1	-0.019	2×10^{-23}	0.000	1	.
CD83	6	14.12	5.9	-0.018	2×10^{-21}	0.000	0.9	.
ANTXR2	4	79.90	6.2	-0.017	6×10^{-20}	-0.001	0.6	***FGF5*, *PRDM8*, *FGF5***
BLK	8	11.49	7.3	-0.017	2×10^{-19}	-0.005	0.005	+
VPS9D1	16	89.71	5.9	0.016	2×10^{-18}	0.003	0.1	+
TMEM71	8	132.69	5.7	-0.016	2×10^{-18}	-0.003	0.1	.
LGALS3BP	17	78.97	6.2	0.016	5×10^{-18}	0.002	0.2	.
GBP2	1	89.11	5.9	0.016	9×10^{-18}	-0.002	0.2	+
PBXIP1	1	154.94	6.6	-0.016	1×10^{-16}	-0.002	0.3	.
PRSS23	11	86.79	6.5	-0.015	6×10^{-16}	0.004	0.03	.
APCDD1	18	10.45	5.8	-0.015	4×10^{-15}	0.000	0.9	.
CASP4LP	11	104.90	5.7	0.015	6×10^{-15}	0.004	0.03	.
ARHGEF40	14	21.07	7.0	-0.015	1×10^{-14}	0.000	0.9	+
TMX4	20	7.98	6.3	-0.015	1×10^{-14}	0.000	0.8	.
MTSS1	8	124.55	5.7	-0.014	3×10^{-14}	0.003	0.07	+
SLC20A1	2	112.65	7.4	0.014	3×10^{-14}	-0.001	0.7	.
SOX8	16	0.98	5.9	0.015	4×10^{-14}	0.000	0.9	.
SLC22A17	14	23.35	5.8	0.014	2×10^{-13}	0.002	0.3	.
IQGAP2	5	76.40	5.7	-0.014	3×10^{-13}	0.003	0.08	.
AXIN2	17	65.53	6.3	0.014	3×10^{-13}	0.001	0.5	.
TPPP3	16	67.39	5.2	-0.014	4×10^{-13}	0.000	1	.
VNN3P	6	132.72	5.8	-0.013	5×10^{-13}	0.002	0.3	+
SORT1	1	109.31	6.4	0.014	7×10^{-13}	-0.004	0.02	+
GABARAPL1	12	10.21	5.2	-0.013	1×10^{-12}	-0.001	0.7	+
STAT1	2	190.91	6.6	0.013	1×10^{-11}	0.001	0.4	+
COCH	14	30.87	5.8	-0.013	1×10^{-11}	0.003	0.2	.
P2RY14	3	151.21	5.6	0.013	1×10^{-11}	-0.004	0.05	.
EBF1	5	158.70	5.6	-0.012	4×10^{-11}	0.006	0.002	+
TXK	4	48.07	7.3	0.012	6×10^{-11}	0.001	0.8	.

Table S6. Additional putative core genes defined by an association $P < 10^{-5}$ and effective *trans*-eQTLs contributing to the eGATE score not less than five but less than eight. *(continued)*

Gene	Transcription site		*trans* score			*cis* score		Reported GWAS hit
	Chr	Start position (Mb)	Effective number of *trans*-eQTLs	Slope	P value	Slope	P value	
ZCCHC14	16	87.41	8.0	0.012	1×10^{-10}	0.002	0.2	***C16orf95***
NSMAF	8	58.58	5.1	-0.012	2×10^{-10}	0.002	0.2	**.**
SYT17	16	19.17	5.9	-0.012	2×10^{-10}	0.000	0.8	**.**
CYB561	17	63.43	5.9	0.012	2×10^{-10}	0.000	1	**+**
FCRL5	1	157.51	6.0	-0.012	4×10^{-10}	0.002	0.3	**+**
ADAM15	1	155.05	5.5	-0.012	5×10^{-10}	0.006	0.001	**+**
DPP4	2	161.99	6.6	0.012	6×10^{-10}	-0.004	0.05	**.**
EPHA1	7	143.39	5.9	0.011	6×10^{-10}	0.001	0.6	**+**
PLAUR	19	43.65	7.5	-0.011	1×10^{-9}	0.004	0.04	**+**
GPER1	7	1.08	7.0	0.011	9×10^{-9}	0.009	9×10^{-7}	**+**
CTTN	11	70.40	5.8	0.011	1×10^{-8}	0.002	0.3	**.**
ITGA5	12	54.40	6.4	-0.011	1×10^{-8}	-0.001	0.6	**.**
FAR2	12	29.15	6.1	-0.010	2×10^{-8}	0.000	0.8	**.**
PRKCA	17	66.30	6.7	0.010	4×10^{-8}	0.000	0.9	**+**
PACSIN2	22	42.84	5.9	-0.010	8×10^{-8}	-0.001	0.8	**.**
GRAMD1C	3	113.83	5.4	-0.010	9×10^{-8}	0.003	0.06	**.**
CYP4F22	19	15.51	7.4	-0.010	1×10^{-7}	-0.002	0.4	**+**
KCNJ2	17	70.17	7.4	0.010	1×10^{-7}	-0.001	0.5	**.**
RAB3IL1	11	61.90	6.3	0.010	2×10^{-7}	0.001	0.6	**.**
MAPRE3	2	26.97	5.8	-0.010	2×10^{-7}	0.004	0.03	**.**
CTNNA1	5	138.61	5.1	-0.010	3×10^{-7}	-0.003	0.07	**.**
NCAM1	11	112.96	6.5	0.009	6×10^{-7}	0.001	0.4	**.**
CEBPE	14	23.12	5.8	0.009	6×10^{-7}	0.004	0.02	***LOC101926933***
PATL2	15	44.67	6.4	0.009	8×10^{-7}	-0.002	0.3	**.**
TSPAN2	1	115.05	5.2	-0.009	1×10^{-6}	0.000	0.8	**+**
ZMYND15	17	4.74	6.6	-0.009	1×10^{-6}	0.005	0.01	**+**
OXER1	2	42.76	6.6	0.009	1×10^{-6}	-0.002	0.4	***HAAO***
IL1R2	2	101.99	5.5	-0.009	2×10^{-6}	0.000	0.8	**+**
MARF1	16	15.59	6.3	0.009	2×10^{-6}	0.002	0.3	**.**
SQLE	8	125.00	5.6	-0.009	2×10^{-6}	0.001	0.7	**.**
GSN	9	121.21	7.0	0.009	2×10^{-6}	-0.001	0.8	**.**
MS4A4A	11	60.19	8.0	-0.009	2×10^{-6}	0.002	0.4	***MS4A6A***
CKB	14	103.52	7.1	-0.009	2×10^{-6}	0.006	0.002	**+**

Table S6. Additional putative core genes defined by an association $P < 10^{-5}$ and effective *trans*-eQTLs contributing to the eGATE score not less than five but less than eight. *(continued)*

Gene	Transcription site		*trans* score			*cis* score		
	Chr	Start position (Mb)	Effective number of *trans*-eQTLs	Slope	P value	Slope	P value	Reported GWAS hit
LGALS1	22	37.68	7.2	-0.009	3×10^{-6}	-0.001	0.5	+
EPHX2	8	27.49	6.7	0.009	3×10^{-6}	0.001	0.6	.
IL1RAP	3	190.51	5.4	-0.009	3×10^{-6}	-0.001	0.4	+
LDB2	4	16.50	7.1	0.009	4×10^{-6}	0.003	0.1	.
MAPK14	6	36.03	5.5	0.009	4×10^{-6}	0.002	0.3	.
OSBPL10	3	31.66	7.1	-0.009	6×10^{-6}	0.000	0.9	.
MS4A1	11	60.46	6.9	-0.009	6×10^{-6}	0.004	0.05	.
FRAT1	10	97.32	5.1	-0.008	9×10^{-6}	0.001	0.8	.

Chr, chromosome; Mb, megabase; Slope, regression coefficient representing the change in MAP (mmHg) per SD increase in the eGATE score. For the analysis 212,569 individuals were included. Reported GWAS hits for essential hypertension are those falling within 200 kb of the transcription start site; NR indicates that the SNP has not been attributed to a specific gene.

Table S7. Additional putative core genes defined by an association $P < 10^{-6}$ and effective *trans*-pQTLs contributing to the pGATE score not less than five but less than twenty.

		Transcription site		*trans* score			*cis* score		
pQTL study	Gene	Chr	Start position (Mb)	Effective number of *trans*-pQTLs	Slope	P value	Slope	P value	Reported GWAS hit
UKB	*LPL*	8	19.90	16.9	-0.021	4×10^{-28}	0.001	0.5	.
UKB	*TIMP4*	3	12.15	16.5	-0.022	6×10^{-28}	0.002	0.4	.
UKB	*C9*	5	39.28	13.1	-0.017	2×10^{-18}	0.002	0.4	+
DeC	*ADIPOQ*	3	186.84	16.3	-0.016	1×10^{-17}	0.003	0.1	+
DeC	*AMBP*	9	114.06	6.9	0.014	5×10^{-15}	0.002	0.4	.
UKB	*EFCAB14*	1	46.67	10.1	0.014	1×10^{-12}	-0.006	0.004	.
UKB	*POMC*	2	25.16	12.3	-0.014	9×10^{-12}	-0.001	0.5	+
UKB	*DCN*	12	91.14	7.0	-0.013	5×10^{-11}	-0.004	0.02	+
DeC	*TFF3*	21	42.31	8.7	0.012	2×10^{-10}	0.000	0.8	+
UKB	*PON3*	7	95.36	18.3	-0.012	2×10^{-10}	0.000	0.9	+
UKB	*NAGPA*	16	5.02	12.0	0.012	2×10^{-10}	-0.001	0.7	.
UKB	*RNASE4*	14	20.68	13.1	0.013	3×10^{-10}	-0.001	0.5	+
UKB	*PNLIPRP1*	10	116.59	15.6	-0.012	4×10^{-10}	0.004	0.07	+
DeC	*SERPINA4*	14	94.56	9.5	0.011	1×10^{-9}	0.001	0.6	+
UKB	*CCL23*	17	36.01	14.0	0.012	1×10^{-9}	0.006	0.003	+
UKB	*SERPINF2*	17	1.74	5.3	0.012	2×10^{-9}	0.001	0.5	+
UKB	*IGSF9*	1	159.93	7.7	0.012	3×10^{-9}	0.002	0.3	.
UKB	*CD163*	12	7.47	12.4	0.012	5×10^{-9}	0.000	0.9	+
DeC	*CADM2*	3	84.96	5.4	-0.011	7×10^{-9}	.	.	.
DeC	*TIMP4*	3	12.15	9.5	-0.011	7×10^{-9}	-0.002	0.2	.
UKB	*F12*	5	177.40	11.7	0.011	1×10^{-8}	0.000	0.8	+
UKB	*ADAMTSL4*	1	150.55	7.8	0.012	1×10^{-8}	0.003	0.2	.
UKB	*HJV*	1	146.02	5.4	0.011	1×10^{-8}	0.003	0.2	+
UKB	*FSTL1*	3	120.39	18.0	-0.011	1×10^{-8}	0.000	0.9	.
DeC	*ALDOB*	9	101.42	8.1	0.011	1×10^{-8}	0.002	0.2	.
UKB	*APCS*	1	159.59	9.1	0.011	2×10^{-8}	-0.001	0.7	.
UKB	*PPY*	17	43.94	17.0	-0.011	2×10^{-8}	.	.	.
UKB	*THY1*	11	119.42	9.4	-0.011	2×10^{-8}	-0.001	0.8	.
UKB	*CLMP*	11	123.07	19.1	-0.011	3×10^{-8}	0.002	0.2	+
UKB	*MSLN*	16	0.76	10.4	0.011	3×10^{-8}	0.005	0.008	.
UKB	*PTH*	11	13.49	14.9	0.011	4×10^{-8}	.	.	.
UKB	*IGFBP1*	7	45.89	6.9	0.011	4×10^{-8}	0.003	0.1	***IGFBP3***
DeC	*PTPRS*	19	5.16	11.9	-0.010	5×10^{-8}	-0.002	0.3	+
UKB	*ISM1*	20	13.22	16.1	-0.011	5×10^{-8}	0.003	0.2	.
UKB	*LRPAP1*	4	3.50	19.5	-0.011	9×10^{-8}	0.001	0.5	+
UKB	*ANGPTL3*	1	62.60	10.8	-0.010	1×10^{-7}	-0.003	0.1	.
UKB	*CPB1*	3	148.79	19.2	-0.010	1×10^{-7}	0.002	0.3	+
UKB	*MYL3*	3	46.84	18.4	-0.011	1×10^{-7}	.	.	.
DeC	*ACAT1*	11	108.12	8.6	0.011	1×10^{-7}	0.002	0.3	.
DeC	*FER*	5	108.75	6.6	-0.010	2×10^{-7}	-0.002	0.2	+
UKB	*SCARB2*	4	76.16	14.6	0.010	2×10^{-7}	0.002	0.2	+
UKB	*BTD*	3	15.60	11.6	0.010	2×10^{-7}	-0.001	0.6	+
DeC	*C5*	9	120.93	9.9	0.010	2×10^{-7}	-0.001	0.6	.
UKB	*CLEC3B*	3	45.00	10.0	-0.010	2×10^{-7}	0.000	1	.
DeC	*TGFBI*	5	136.03	7.6	0.010	2×10^{-7}	0.001	0.5	+

Table S7. Additional putative core genes defined by an association $P < 10^{-6}$ and effective *trans*-pQTLs contributing to the pGATE score not less than five but less than twenty. *(continued)*

		Transcription site		*trans* score			*cis* score		
pQTL study	Gene	Chr	Start position (Mb)	Effective number of *trans*-pQTLs	Slope	P value	Slope	P value	Reported GWAS hit
UKB	*CFI*	4	109.73	6.9	0.010	3×10^{-7}	0.000	1	.
UKB	*NRP1*	10	33.18	6.8	0.010	3×10^{-7}	-0.001	0.7	.
DeC	*NPPB*	1	11.86	7.1	-0.010	4×10^{-7}	-0.015	5×10^{-15}	***MTHFR***
UKB	*C2*	6	31.90	8.3	0.010	4×10^{-7}	.	.	***HCG2248***, ***LINC02571***, ***HCP5***, ***BAT2***, ***BAT5***, ***MPIG6B***, ***SLC44A4***, ***CYP21A2***
UKB	*ERBB3*	12	56.08	16.6	0.010	5×10^{-7}	0.006	0.004	.
UKB	*ADAM23*	2	206.44	11.3	0.010	5×10^{-7}	-0.003	0.2	.
UKB	*RARRES2*	7	150.34	6.5	0.010	5×10^{-7}	0.002	0.3	.
UKB	*ATRN*	20	3.47	10.8	0.010	7×10^{-7}	0.001	0.5	.
UKB	*SOD3*	4	24.79	13.9	-0.010	7×10^{-7}	0.003	0.2	.
UKB	*FST*	5	53.48	7.2	0.010	8×10^{-7}	-0.001	0.7	+
UKB	*SPOCK1*	5	136.98	14.5	-0.010	1×10^{-6}	-0.002	0.2	.
INT	*JAK2*	9	4.98	5.5	-0.009	1×10^{-6}	.	.	.
DeC	*KNG1*	3	186.72	6.5	0.009	1×10^{-6}	0.001	0.5	+
DeC	*AGT*	1	230.69	8.8	0.009	1×10^{-6}	0.004	0.04	+
INT	*PPY*	17	43.94	6.6	-0.009	2×10^{-6}	.	.	.
DeC	*AKR1D1*	7	138.00	5.3	0.009	2×10^{-6}	0.000	1	.
DeC	*LEAP2*	5	132.87	7.4	0.009	2×10^{-6}	-0.003	0.08	.
DeC	*HS6ST2*	X	132.63	9.8	-0.009	2×10^{-6}	.	.	.
DeC	*PZP*	12	9.15	14.7	0.009	2×10^{-6}	0.001	0.8	+
UKB	*NPPB*	1	11.86	16.3	-0.009	2×10^{-6}	-0.020	5×10^{-23}	***MTHFR***
DeC	*IL2*	4	122.45	6.0	0.009	2×10^{-6}	.	.	.
UKB	*NEFL*	8	24.95	12.9	0.009	2×10^{-6}	.	.	.
DeC	*GNPTG*	16	1.35	6.6	0.009	3×10^{-6}	-0.003	0.2	+
UKB	*LECT2*	5	135.92	8.8	0.009	3×10^{-6}	0.000	0.8	+
UKB	*CDH3*	16	68.64	16.3	-0.009	4×10^{-6}	0.004	0.06	.
UKB	*SNX15*	11	65.03	5.9	-0.009	4×10^{-6}	-0.003	0.1	+
UKB	*FCGR2A*	1	161.51	11.3	0.009	4×10^{-6}	-0.003	0.1	+
DeC	*CTSE*	1	206.01	6.3	-0.009	5×10^{-6}	-0.004	0.04	.
UKB	*ACRV1*	11	125.67	6.9	0.009	5×10^{-6}	0.003	0.08	.
UKB	*TNFRSF6B*	20	63.70	6.3	0.009	5×10^{-6}	-0.004	0.05	+
DeC	*MMP1*	11	102.79	18.5	-0.008	6×10^{-6}	0.002	0.2	+
UKB	*EPHA2*	1	16.12	18.9	-0.009	6×10^{-6}	-0.001	0.6	.
UKB	*SCARA5*	8	27.87	9.6	-0.009	7×10^{-6}	0.000	0.9	+
UKB	*C3*	19	6.68	6.3	0.009	8×10^{-6}	0.001	0.5	+
UKB	*CXCL12*	10	44.37	18.0	-0.009	8×10^{-6}	-0.003	0.1	.
DeC	*FCN3*	1	27.37	13.0	0.008	8×10^{-6}	0.003	0.1	+

Table S7. Additional putative core genes defined by an association $P < 10^{-6}$ and effective *trans*-pQTLs contributing to the pGATE score not less than five but less than twenty. *(continued)*

		Transcription site		*trans* score			*cis* score		
pQTL study	Gene	Chr	Start position (Mb)	Effective number of *trans*-pQTLs	Slope	P value	Slope	P value	Reported GWAS hit
DeC	*EHMT2*	6	31.88	10.7	-0.008	8×10^{-6}	.	.	***HCG2248*, *LINC02571*, *HCP5*, *BAT2*, *BAT5*, *MPIG6B*, *SLC44A4*, *CYP21A2***
UKB	*CRNN*	1	152.41	8.7	-0.009	9×10^{-6}	0.003	0.07	**.**
DeC	*AKR7A3*	1	19.28	6.4	0.008	1×10^{-5}	-0.001	0.5	**+**
DeC	*CPB1*	3	148.79	13.0	-0.008	1×10^{-5}	0.001	0.7	**+**
DeC	*UBLCP1*	5	159.26	8.0	0.008	1×10^{-5}	-0.001	0.6	**.**

DeC, deCODE study; UKB, UK Biobank Pharma Proteomics Project; Chr, chromosome; Mb, megabase; Slope, regression coefficient representing the change in MAP (mmHg) per SD increase in the pGATE score. For the analysis 212,569 individuals were included from DeC and 190,635 individuals from UKB, with participants enrolled in the proteomics sub-study excluded. Reported GWAS hits for essential hypertension are those falling within 200 kb of the transcription start site; NR indicates that the SNP has not been attributed to a specific gene. Where multiple pGATE scores exist for a given protein (e.g., derived from different pQTL GWAS), only the score with the strongest association is reported.